



# Robotics, VIMA, Pi, Gemini-Robotics

Presenter: Zhihan Wang

| Work (year)                                                                   | Problem / task focus                                                                       | Main contribution                                                                                                                                                                                   |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Natural Language Communication with Robots</b> (≈2016; blocks-world style) | Ground free-form human language to concrete manipulation plans in a simplified world       | Shows that <b>unrestricted language</b> (not templates) can specify spatially complex goals; introduces datasets/evals for language→action grounding                                                |
| <b>VIMA</b> (2023)                                                            | General robot manipulation via <b>multimodal prompting</b> ; test zero-shot generalization | Reframes tasking as <b>interleaved text-and-image prompts</b> ; creates <b>VIMA-Bench</b> (L1→L4) for comp-gen generalization                                                                       |
| <b>π0 (pi-zero)</b> (2024/2025)                                               | <b>Reactive, fluent control</b> for diverse real-robot tasks across embodiments            | Introduces <b>continuous action distribution</b> via <b>conditional flow matching</b> and <b>action chunks</b> for smooth high-rate control; large cross-embodiment pretrain + dexterous post-train |

# Natural Language Communication with Robots

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“bridging the communication gap between humans and robots”



$t_0$ : Initial state



$t_8$



$t_9$



$t_{20}$ : Final drawing

**Figure 1:** Above are four states in a 20-action sequence for drawing the digit 5. The world state can be encoded for the learner as IDs and locations or raw images. Annotations are provided between every adjacent state (See Table 1) or between sequences (e.g.  $t_8 \rightarrow t_{20}$ : *Create a four-line diagonal which moves to the southeast. Starting with Heineken ...* ) to describe multi-action plans.

# Data

Starting from the shape and move to random position, mturk for labeling.

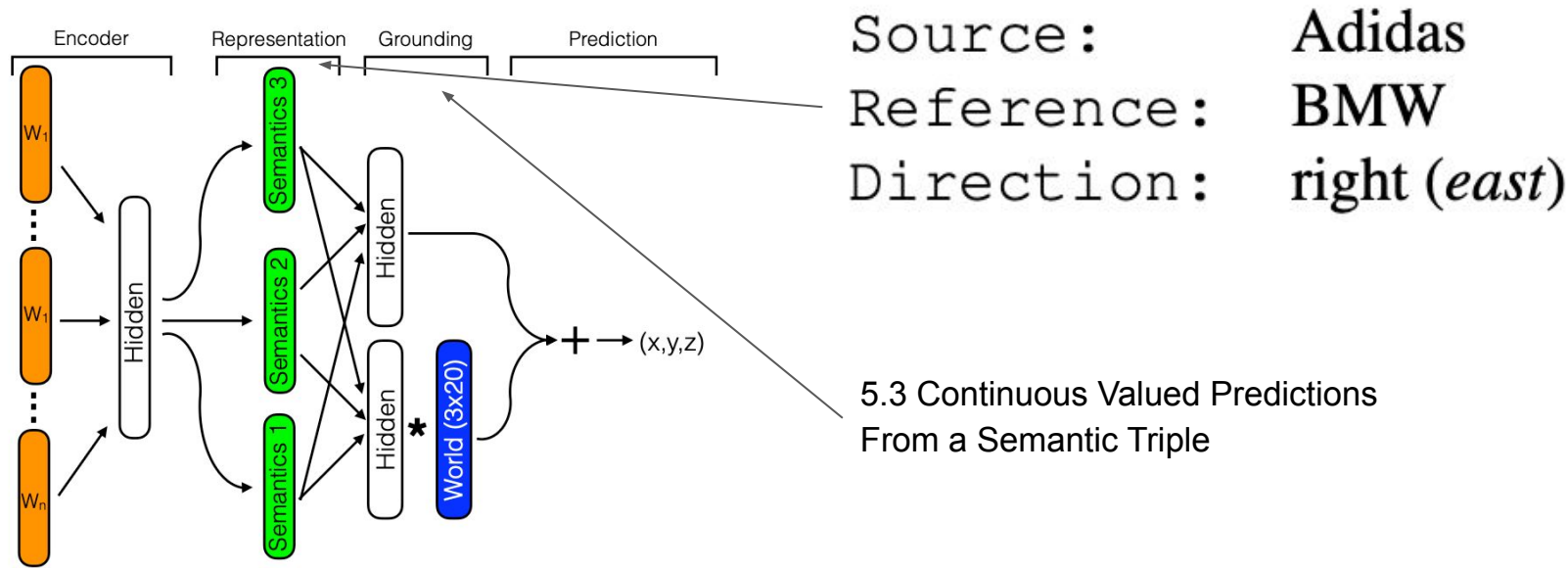
# Learning Problems of Interest

**Entity Grounding:** “there are many ways an object might be referred to in everyday speech”

**Spatial Relation Grounding:** “coca cola, hp, nvidia”

**Plan Recognition:** *slide the adidas block 2 blocks straight up . then  
slide it 6 block spaces to the left .*

The models we introduce in this paper have difficulty dealing with these kind of commands.



**Figure 2:** Our models all follow the above architecture. 1-Hot word vectors (orange) are fed as input to a Feed-Forward or Recurrent Neural Network for encoding. A semantic representation is extracted (green), which in conjunction with knowledge of the world (blue) is grounded to predict an action.

|                   |                        | MNIST Patterns with labeled blocks |             |             |             |     |     |     | Random Patterns with blank blocks |             |             |             |     |     |     |
|-------------------|------------------------|------------------------------------|-------------|-------------|-------------|-----|-----|-----|-----------------------------------|-------------|-------------|-------------|-----|-----|-----|
|                   |                        | Source                             |             | Target      |             |     |     |     | Source                            |             | Target      |             |     |     |     |
|                   |                        | Med                                | Mean        | Med         | Mean        | S   | R   | D   | Med                               | Mean        | Med         | Mean        | S   | R   | D   |
| Human Performance |                        | 0.00                               | 0.00        | 0.21        | 0.53        | 100 |     |     | 0.00                              | 0.30        | 0.37        | 1.39        | 93  |     |     |
| Oracle            |                        | –                                  | –           | 0.00        | 0.45        | 100 | 100 | 100 | –                                 | –           | 1.00        | 1.09        | 100 | 100 | 100 |
| FFN               | Discrete Predictions   | <b>0.00</b>                        | 0.49        | <b>1.09</b> | 2.17        | 93  | 69  | 63  | 5.28                              | 5.09        | 5.51        | 5.46        | 9   | 15  | 32  |
|                   | Continuous Predictions | 0.49                               | 1.00        | 1.59        | 2.42        |     |     |     | 4.25                              | 4.04        | 3.86        | <b>3.93</b> |     |     |     |
|                   | End-to-End             | 0.02                               | <b>0.38</b> | 1.14        | <b>1.81</b> |     |     |     | <b>3.45</b>                       | <b>3.52</b> | <b>3.60</b> | 3.94        |     |     |     |
| RNN               | Discrete Predictions   | <b>0.00</b>                        | <b>0.14</b> | <b>0.00</b> | <b>0.98</b> | 98  | 92  | 78  | 5.29                              | 5.00        | 5.51        | 5.57        | 10  | 7   | 46  |
|                   | Continuous Predictions | 0.47                               | 0.64        | 1.23        | 1.60        |     |     |     | 4.16                              | 4.05        | 3.71        | 3.87        |     |     |     |
|                   | End-to-End             | 0.03                               | 0.19        | 0.53        | 1.05        |     |     |     | <b>3.29</b>                       | <b>3.47</b> | <b>3.60</b> | <b>3.70</b> |     |     |     |
| Center Baseline   |                        | –                                  | –           | 3.46        | 3.43        | 100 |     |     | –                                 | –           | 4.09        | 4.06        | 100 |     |     |
| Random Baseline   |                        | 6.37                               | 6.49        | 6.12        | 6.21        | 5   | 5   | 11  | 4.90                              | 4.97        | 5.51        | 5.44        | 10  | 11  | 12  |

**Table 4:** Model error when trained on only the subset of the data with decorated blocks or blank blocks. Where appropriate S, R, and D are the model’s predictive accuracy at identifying the Source, Reference and Direction. All models are evaluated on the Median and Mean prediction error the source block and its final target location. Distances are presented in block-lengths.



# Error Analysis

| Error Type              | Count | Example                                                                                                                                                                                 |
|-------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Relation Actions  | 20    | Place block 20 parallel with the 8 block and slightly to the right of the 6 block. Place block 15 on the same vertical column as blocks 16 and 17, and two rows above blocks 11 and 3.  |
| Geometric Understanding | 10    | Continue the diagonal row of 20, 19 and 15 downward with 13. Put block 12 in the column between the columns with blocks 4 and 13, and on the same row as the lowest block on the board. |
| Grammatical Ambiguity   | 10    | 19 moved from behind the 8 to under the 18th block.<br>Burger King tile should be directly above the Coca Cola tile. Move Coca Cola.                                                    |
| Grounding Names         | 5     | Put the block that looks like a taurus symbol just above the bird.                                                                                                                      |
| Understanding Distance  | 5     | move the texaco block 5 block lengths above the BMW block                                                                                                                               |

**Table 5:** We performed a subjective error analysis of the results of our Fixed Semantics model using the RNN encoder. Example sentences and the frequency of each type of error are reported above from the worst 50 errors on the development data.

## Main Contribution:

Shows that unrestricted language (not templates) can specify spatially complex goals; introduces datasets/evals for language→action grounding

## Limitations:

Limited physical realism; sparse perception; no low-level reactive control; struggles with ambiguous spatial language

# VIMA: General Robot Manipulation with Multimodal Prompts

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**Li Fei-Fei**<sup>1</sup> **Anima Anandkumar**<sup>3,4</sup> **Yuke Zhu**<sup>3,6‡</sup> **Linxi Fan**<sup>3‡</sup>

“wide spectrum of robot manipulation tasks can be expressed with multimodal prompts”

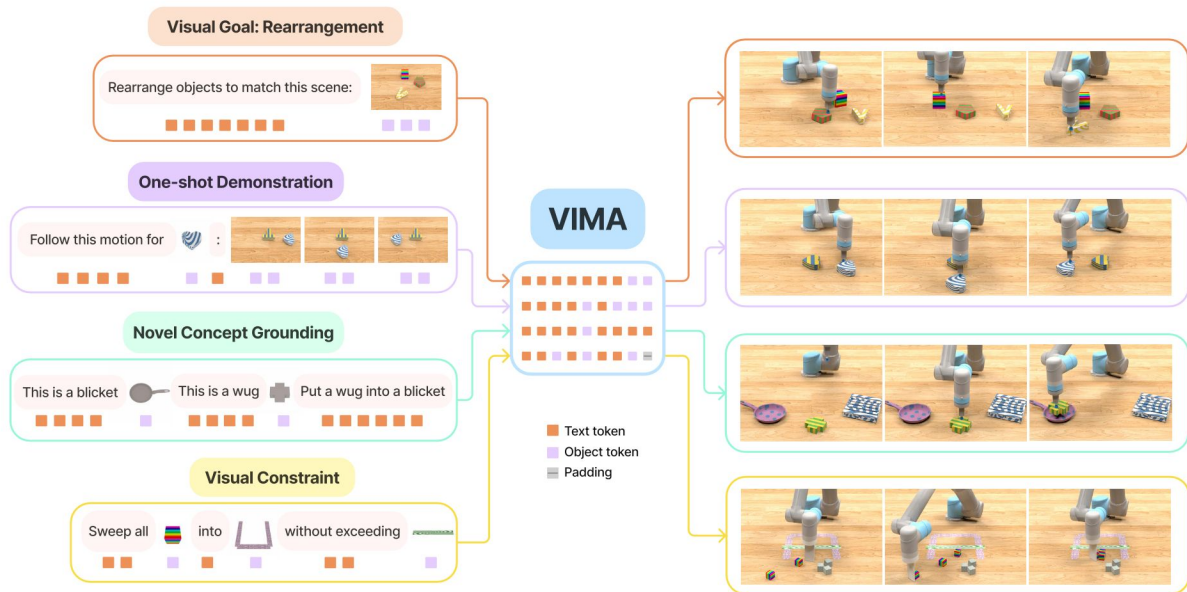


Figure 1: **Multimodal prompts for task specification.** We observe that many robot manipulation tasks can be expressed as *multimodal prompts* that interleave language and image/video frames. We introduce VIMA, an embodied agent capable of processing multimodal prompts (left) and controlling a robot arm to solve the task (right).

# Data

- Env: VIMA-BENCH built on Ravens (PyBullet, UR5, 0.5×1 m table), with extensible 3D objects/textures and RGB (frontal/top-down).
- Training Data: Offline imitation from scripted oracles; ~650k successful trajectories ( $\approx 50$ k/task, 17 tasks). Actions are high-level primitives (pick/place, push/wipe) parameterized by poses.

# Generalization

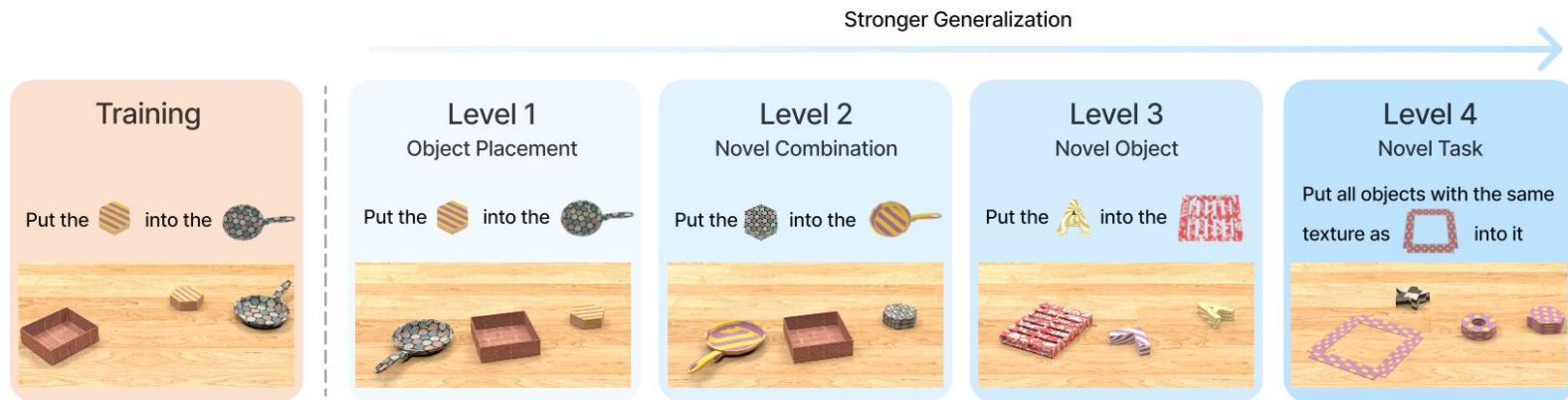


Figure 2: **Evaluation Protocol in VIMA-BENCH.** We design 4 levels of evaluation settings to systematically measure the zero-shot generalization capability of an agent. Each level deviates more from the training distribution, and thus is strictly more challenging than the previous level.

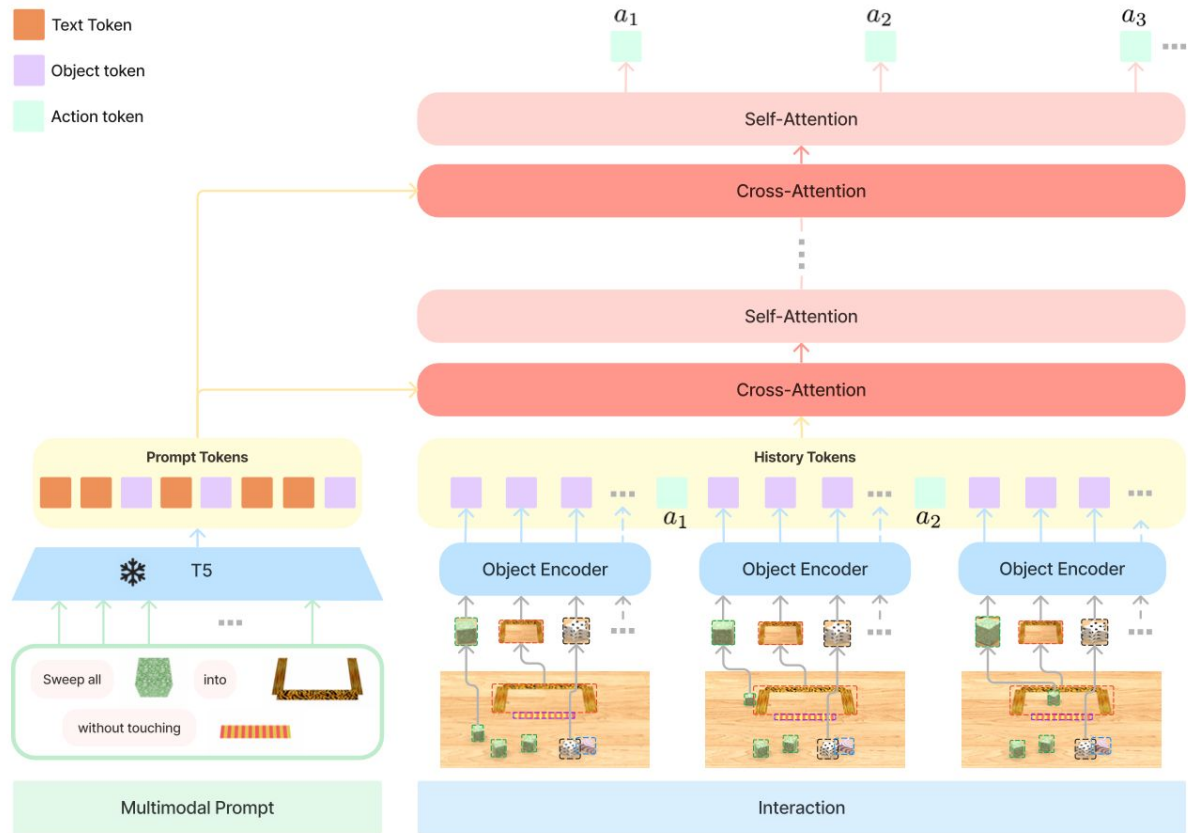


Figure 3: **VIMA Architecture.** We encode the multimodal prompts with a pre-trained T5 model, and condition the robot controller on the prompt through cross-attention layers. The controller is a causal transformer decoder consisting of alternating self and **cross attention** layers that predicts motor commands conditioned on prompts and interaction history.

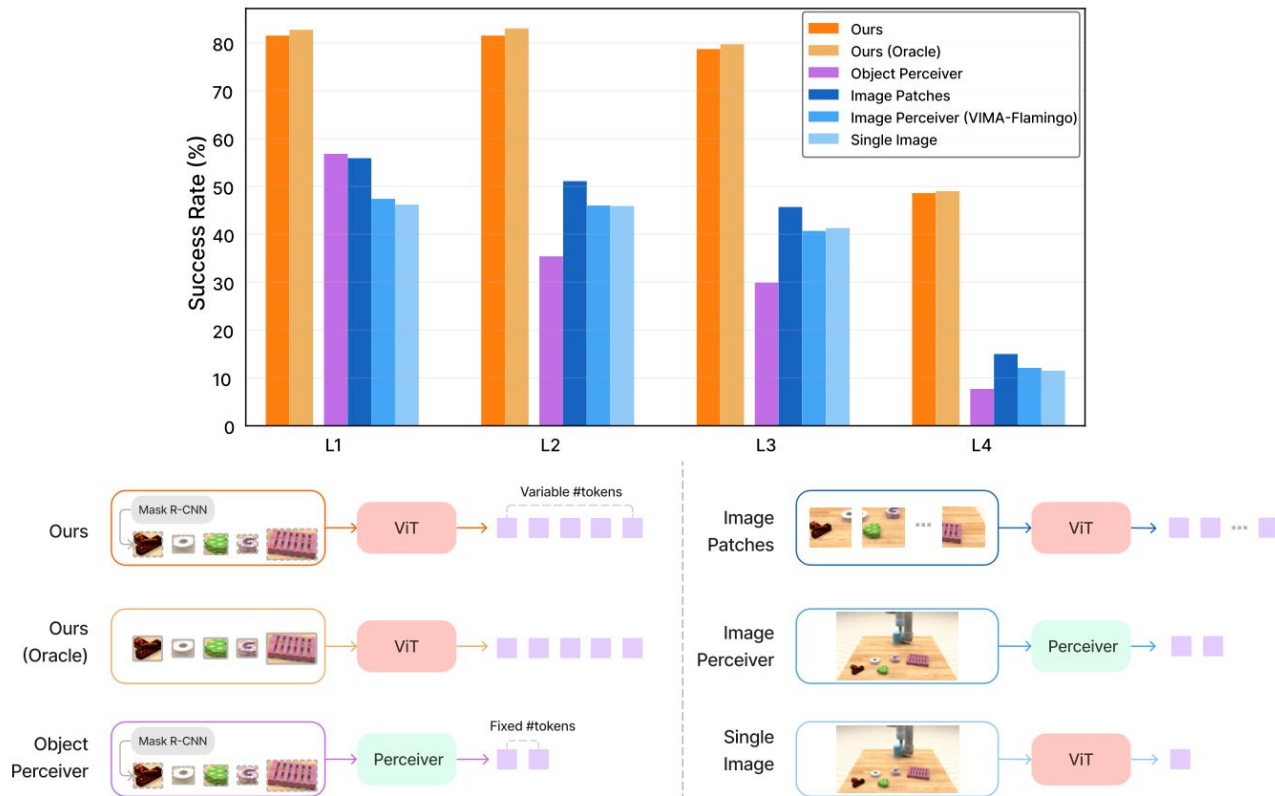


Figure 6: **Ablation on visual tokenizers.** We compare the performance of VIMA-200M model across different visual tokenizers. Our proposed object tokens outperform all methods that learn directly from raw pixels, and *Object Perceiver* that downsamples the object sequence to a fixed number of tokens.

Main Contribution: Reframes tasking as interleaved text-and-image prompts; creates VIMA-Bench (L1→L4) for generalization

Limitations: Mostly simulation; no high-frequency, low-level control; relies on pre-defined primitives; performance depends on prompt quality

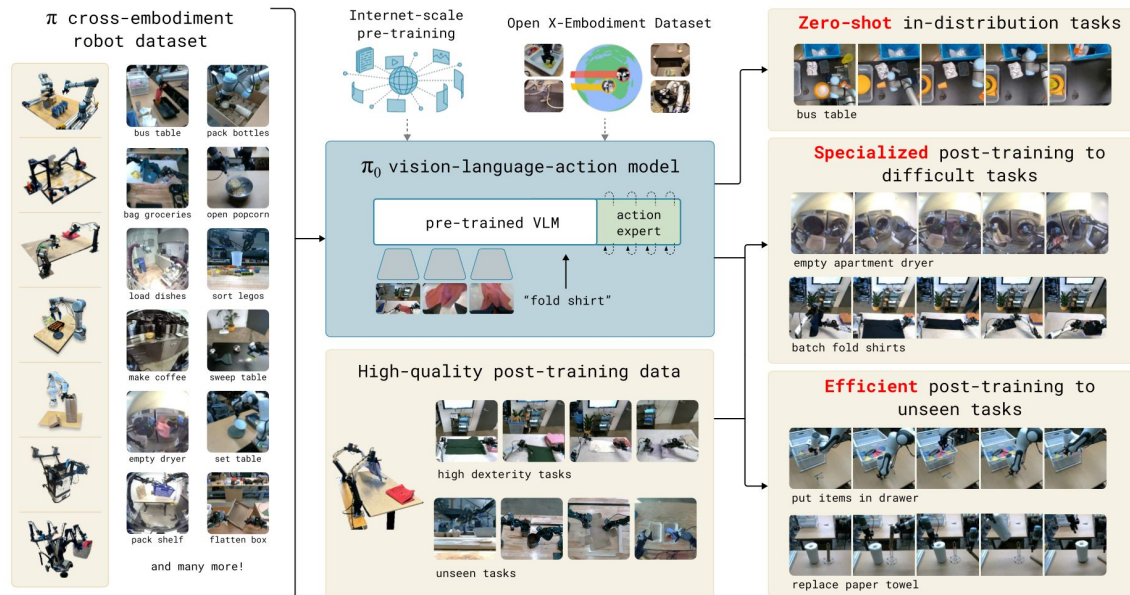


Project Page: <https://vimalabs.github.io/>

# $\pi_0$ : A Vision-Language-Action Flow Model for General Robot Control

## Physical Intelligence

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, Ury Zhilinsky  
<https://physicalintelligence.company/blog/pi0>



# Task

Reactive, fluent control for diverse real-robot tasks across embodiments

Input: RGB (multi-view), language, proprioception; optional task hinting

Output: Continuous action sequences (e.g., 50 Hz action chunks); low-level control over the robot's native DoFs (arms, grippers, etc.)

# Data

- Pretrain Mix: Own dexterous datasets collected on 7 robot configurations across 68 tasks + full OXE (data from 22 robots); ~10,000 hours total. Includes language labels (task names + ~2 s segment annotations).
- Post-training: High-quality curated sets for long-horizon dexterity (e.g., laundry folding, table bussing, box assembly).
- Backbone Init: PaliGemma VLM weights (internet-scale pretraining) before robot data finetuning.

# Action Expert

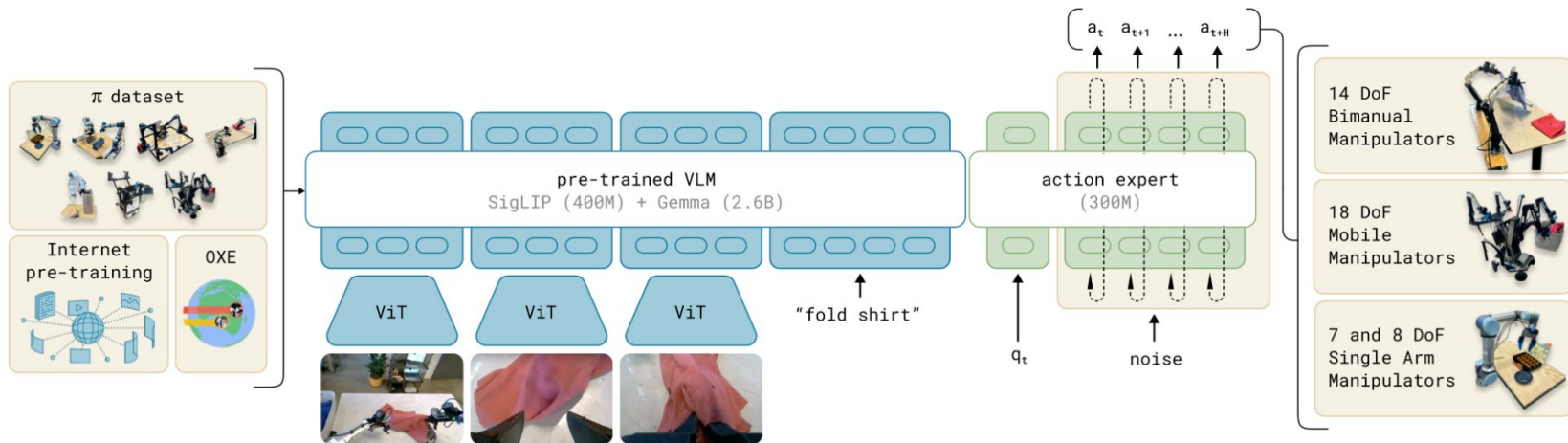


Fig. 3: **Overview of our framework.** We start with a pre-training mixture, which consists of both our own dexterous manipulation datasets and open-source data. We use this mixture to train our flow matching VLA model, which consists of a larger VLM backbone and a smaller *action expert* for processing robot states and actions. The VLM backbone weights are initialized from PaliGemma [5], providing representations learned from large-scale Internet pre-training. The resulting  $\pi_0$  model can be used to control multiple robot embodiments with differing action spaces to accomplish a wide variety of tasks.

# Flow

Formally, we want to model the data distribution  $p(\mathbf{A}_t|\mathbf{o}_t)$ , where  $\mathbf{A}_t = [\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+H-1}]$  corresponds to an *action chunk* of future actions (we use  $H = 50$  for our tasks), and  $\mathbf{o}_t$  is an observation. The observation consists of multiple RGB images, a language command, and the robot’s proprioceptive state, such that  $\mathbf{o}_t = [\mathbf{I}_t^1, \dots, \mathbf{I}_t^n, \ell_t, \mathbf{q}_t]$ , where  $\mathbf{I}_t^i$  is  $i^{\text{th}}$  image (with 2 or 3 images per robot),  $\ell_t$  is a sequence of language tokens, and  $\mathbf{q}_t$  is a vector of joint angles. The images  $\mathbf{I}_t^i$  and state  $\mathbf{q}_t$  are encoded via corresponding encoders and then projected via a linear projection layer into the same embedding space as the language tokens.

For each action  $\mathbf{a}_{t'}$  in the action chunk  $\mathbf{A}_t$ , we have a corresponding *action token* that we feed through the action expert. During training, we supervise these action tokens using a conditional flow matching loss [28, 32],

$$L^\tau(\theta) = \mathbb{E}_{p(\mathbf{A}_t|\mathbf{o}_t), q(\mathbf{A}_t^\tau|\mathbf{A}_t)} \|\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t) - \mathbf{u}(\mathbf{A}_t^\tau|\mathbf{A}_t)\|^2,$$

where subscripts denote robot timesteps and superscripts denote flow matching timesteps, with  $\tau \in [0, 1]$ . Recent work in high-resolution image [14] and video [38] synthesis has shown that flow matching can achieve strong empirical performance when combined with a simple linear-Gaussian (or optimal transport) probability path [28], given by  $q(\mathbf{A}_t^\tau|\mathbf{A}_t) = \mathcal{N}(\tau\mathbf{A}_t, (1-\tau)\mathbf{I})$ . In practice, the network is trained by sampling random noise  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , computing the “noisy actions”  $\mathbf{A}_t^\tau = \tau\mathbf{A}_t + (1-\tau)\epsilon$ , and then training the network outputs  $\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t)$  to match the denoising vector field  $\mathbf{u}(\mathbf{A}_t^\tau|\mathbf{A}_t) = \epsilon - \mathbf{A}_t$ . The action expert uses a full bidirectional attention mask, so that all action tokens attend to each other. During training, we sample the flow matching timestep  $\tau$  from a beta distribution that emphasizes lower (noisier) timesteps. See Appendix B for more details.

At inference time, we generate actions by integrating the learned vector field from  $\tau = 0$  to  $\tau = 1$ , starting with random noise  $\mathbf{A}_t^0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ . We use the forward Euler integration rule:

$$\mathbf{A}_t^{\tau+\delta} = \mathbf{A}_t^\tau + \delta\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t),$$

where  $\delta$  is the integration step size. We use 10 integration steps (corresponding to  $\delta = 0.1$ ) in our experiments. Note that inference can be implemented efficiently by caching the attention keys and values for the prefix  $\mathbf{o}_t$  and only recomputing the suffix corresponding to the action tokens for each integration step. We provide more details regarding the inference procedure, including the inference time for each part of the model, in Appendix D.

# Flow

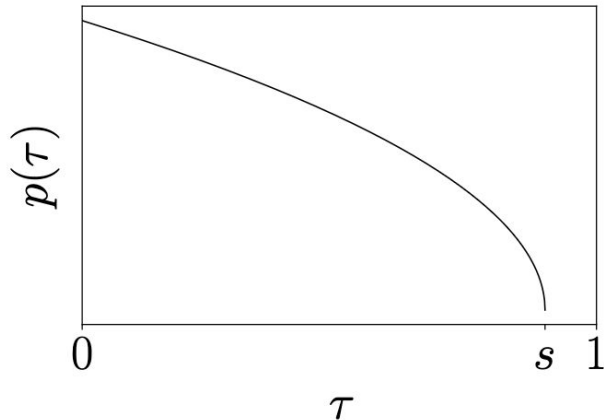


Fig. 14: **Flow matching timestep sampling distribution.** We sample  $\tau$  from a shifted beta distribution that emphasizes lower timesteps (corresponding to noisier actions), and does not sample timesteps at all above a cutoff value  $s$ . We use  $s = 0.999$  in our experiments.

However, we hypothesize that the task of action prediction is subtly different from high-resolution image synthesis — while it may be relatively easy to predict the mean image conditioned on a text label, predicting the mean action conditioned on a robot observation (i.e., learning  $\mathbb{E}[\mathbf{A}_t|\mathbf{o}_t]$ ) is a much harder problem; this is because the observation  $\mathbf{o}_t$  is very *informative* in that it should constrain the distribution of possible actions much more than a text label constrains the distribution of possible images. As a result, we designed a timestep sampling distribution that emphasizes low timesteps (high noise levels); additionally, timesteps above a given threshold  $s$  are not sampled at all, since they are not needed so long as the integration step  $\delta$  is greater than  $1 - s$ . The distribution is given

# A Question

“Flow matching provides our model with high precision and multimodal modeling capability, making it especially well suited to high-frequency dexterous tasks.”

????



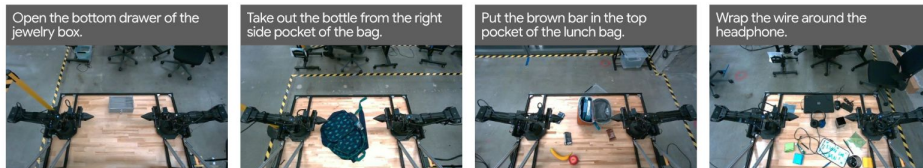
Main Contribution: introduces continuous action distribution via conditional flow matching and action chunks for smooth high-rate control; large cross-embodiment pretrain + dexterous post-train

Limitations: Data-hungry; performance can depend on embodiment mismatch

# Gemini Robotics: Bringing AI into the Physical World

Gemini Robotics Team, Google DeepMind<sup>1</sup>

Dexterous, general & instructable Vision-Language-Action model



Complex dexterous tasks



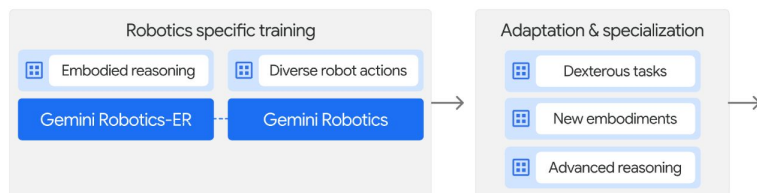
New embodiments



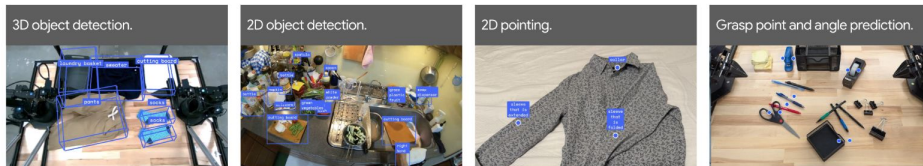
Advanced reasoning & acting



◆ Gemini 2.0 →



Advanced embodied reasoning for robotics



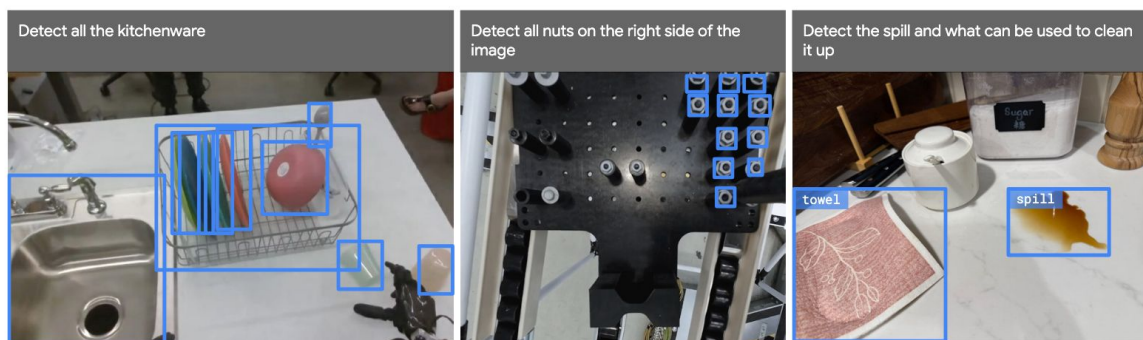
**ERQA:** An open-source benchmark specifically designed to evaluate embodied reasoning capabilities of multimodal models, addressing the lack of benchmarks that go beyond assessing atomic capabilities and facilitating standardized assessment and future research.

| Trajectory Reasoning                                                                                                                                                                                                                                   | Action Reasoning                                                                                                                                                                                                                                                                                  | Spatial Reasoning                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Approximately which colored trajectory should the zipper follow to begin zipping up the suitcase?</p> <p><b>A. Blue</b><br/>B. Purple<br/>C. Green<br/>D. Red</p> |  <p>How should the person move the wrench so that it is ready to rotate the hex screw closest to it?</p> <p><b>A. Forward and right</b><br/>B. Up and left<br/>C. Forward and left<br/>D. None of the above</p> |  <p>There are 4 sinks in the picture. Which arrow points to the one that is closest to the viewer?</p> <p>A. Cyan<br/>B. Blue<br/>C. Red<br/><b>D. None of the arrows</b></p> |

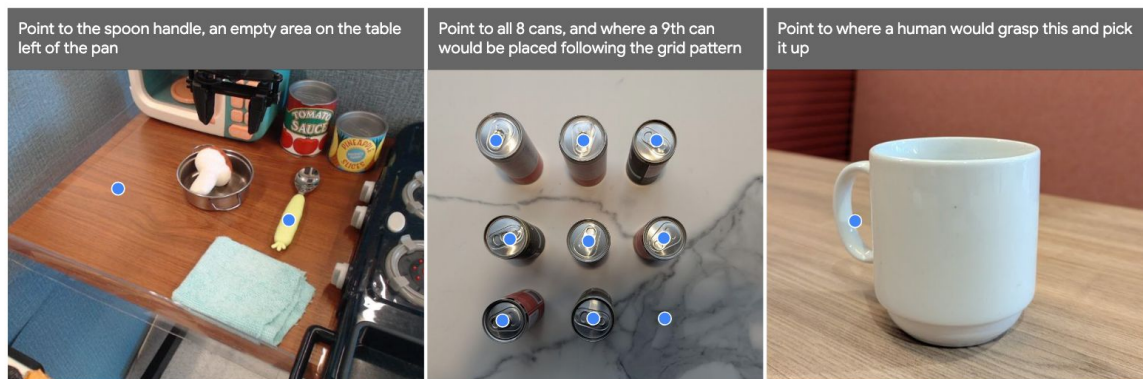
**Figure 3** | Example questions from the Embodied Reasoning Question Answering (ERQA) benchmark, with answers in bold.

Gemini Robotics-ER: A VLM demonstrating enhanced embodied reasoning capabilities.

# Gemini ER



**Figure 6** | 2D Detection examples with Gemini 2.0 Flash. Left: detect by object category. Middle: detect by spatial description. Right: detect by affordance. Predicted object labels are not shown for left and middle images to reduce visual clutter.



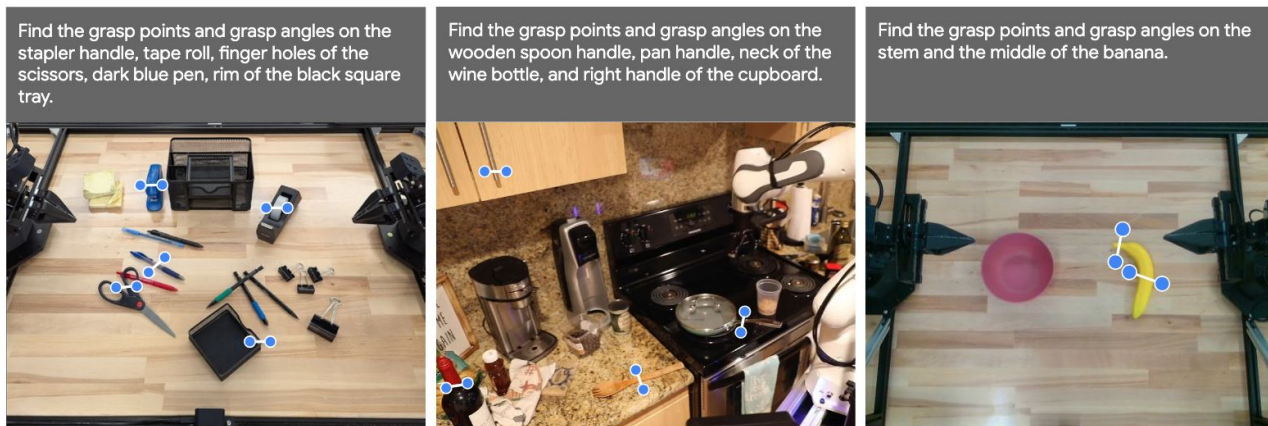
**Figure 7** | Gemini 2.0 can predict 2D points from natural language queries. Examples are obtained with Gemini 2.0 Flash. Predicted point labels are not visualized.



# Gemini ER



**Figure 8** | Gemini 2.0 can predict 2D trajectories by first predicting start and end points. Examples are obtained with Gemini 2.0 Flash. Predicted point labels are not visualized.



**Figure 9** | Gemini Robotics-ER can predict top-down grasps by leveraging Gemini 2.0's 2D pointing capability. Examples are obtained with Gemini Robotics-ER.

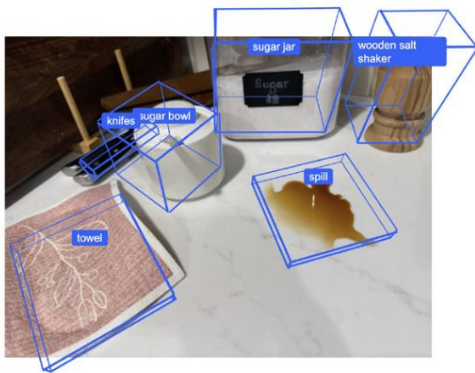
# Gemini ER



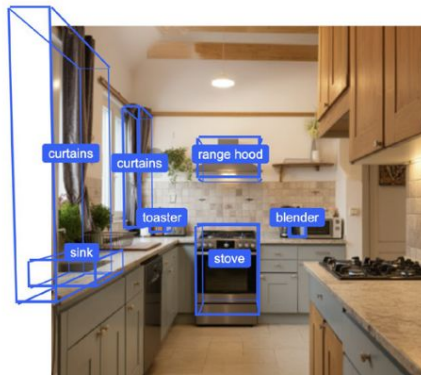
**Figure 10** | Gemini 2.0 can understand 3D scenes by correlating 2D points across different views. For each image pair, the left image with the point coordinates and the right image without coordinates are given, and the model predicts which of the labeled points in the left image are visible in the right image, as well as the coordinates of the visible points in the right image. Examples are obtained with Gemini 2.0 Flash.

# Gemini ER

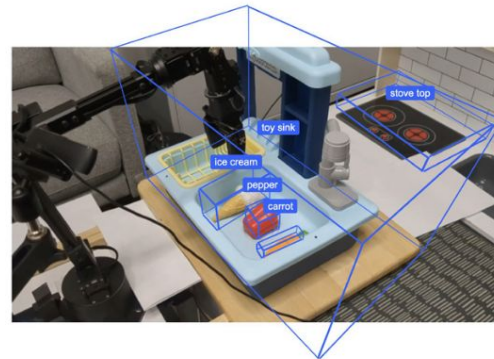
Find the 3D bounding boxes of sugar bowl, sugar jar, wooden salt shaker, knives, spill, towel.



Detect the 3D bounding boxes of blender, toaster, curtains, sink, range hood, stove.



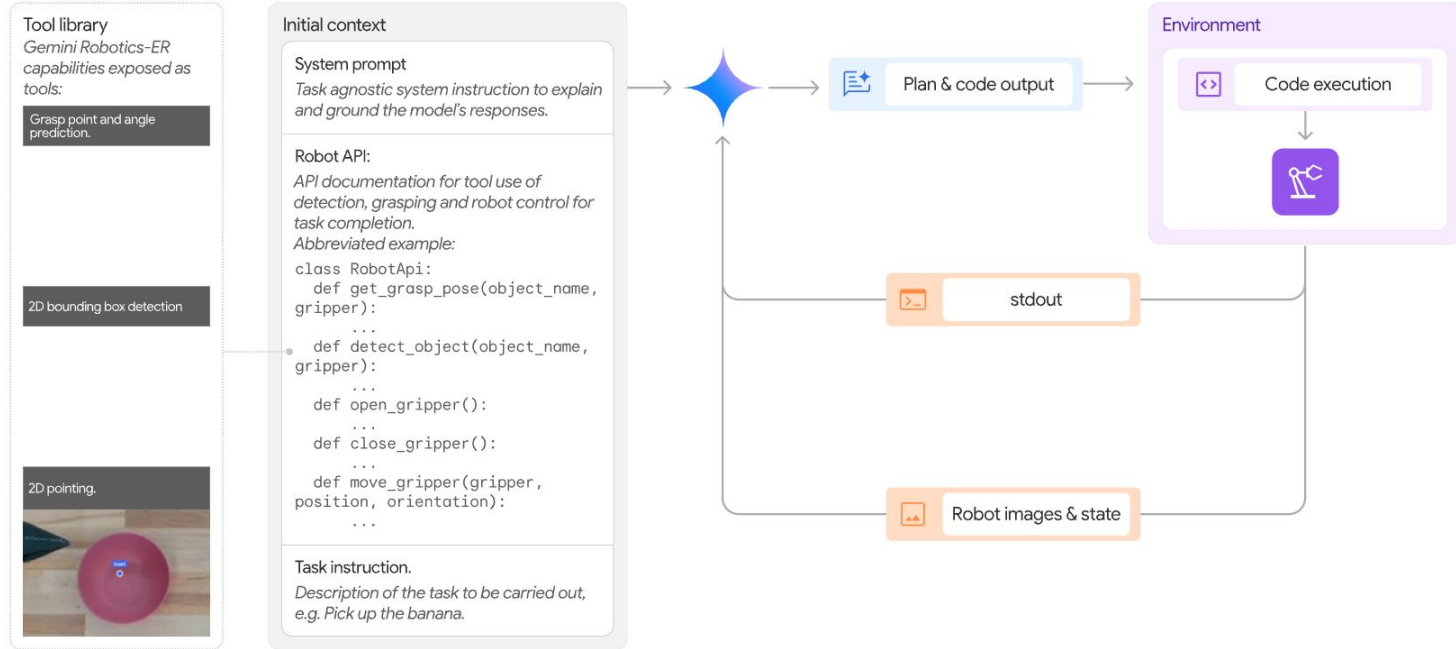
Detect the 3D bounding boxes of stove top, toy sink, ice cream, pepper, carrot.



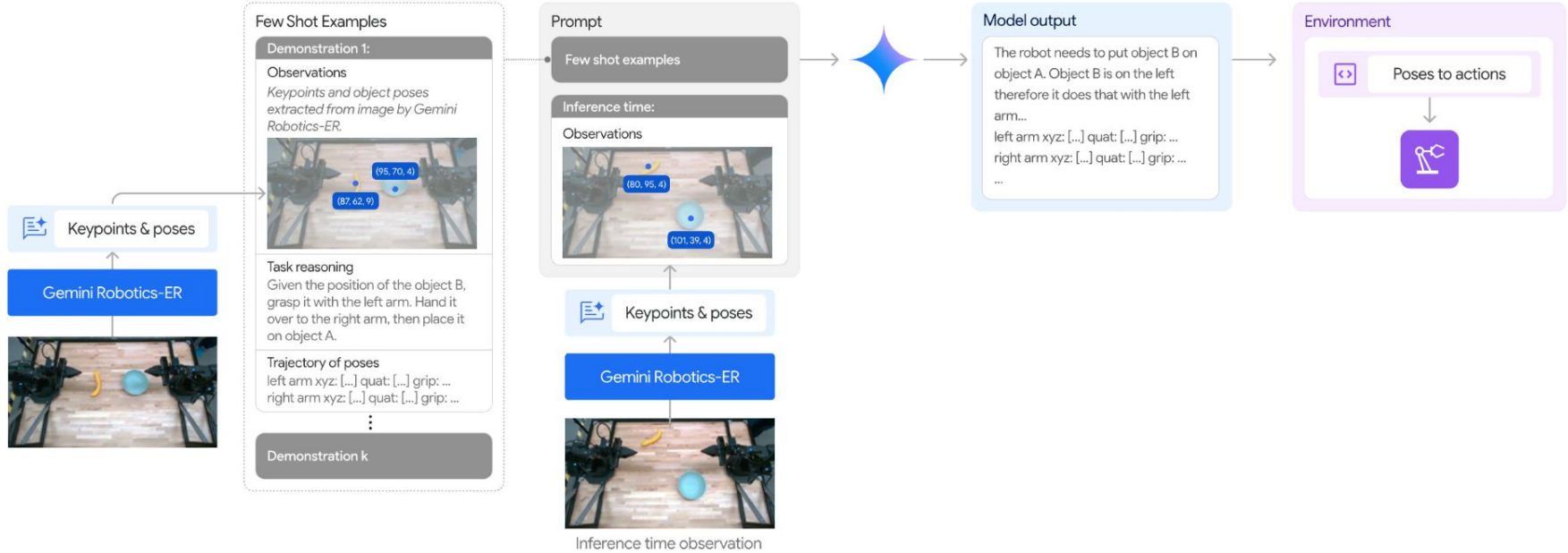
**Figure 11** | Gemini 2.0 can directly predict open-vocabulary 3D object bounding boxes. Examples are obtained with Gemini 2.0 Flash.



# Gemini 2.0 Enables Zero and Few-Shot Robot Control

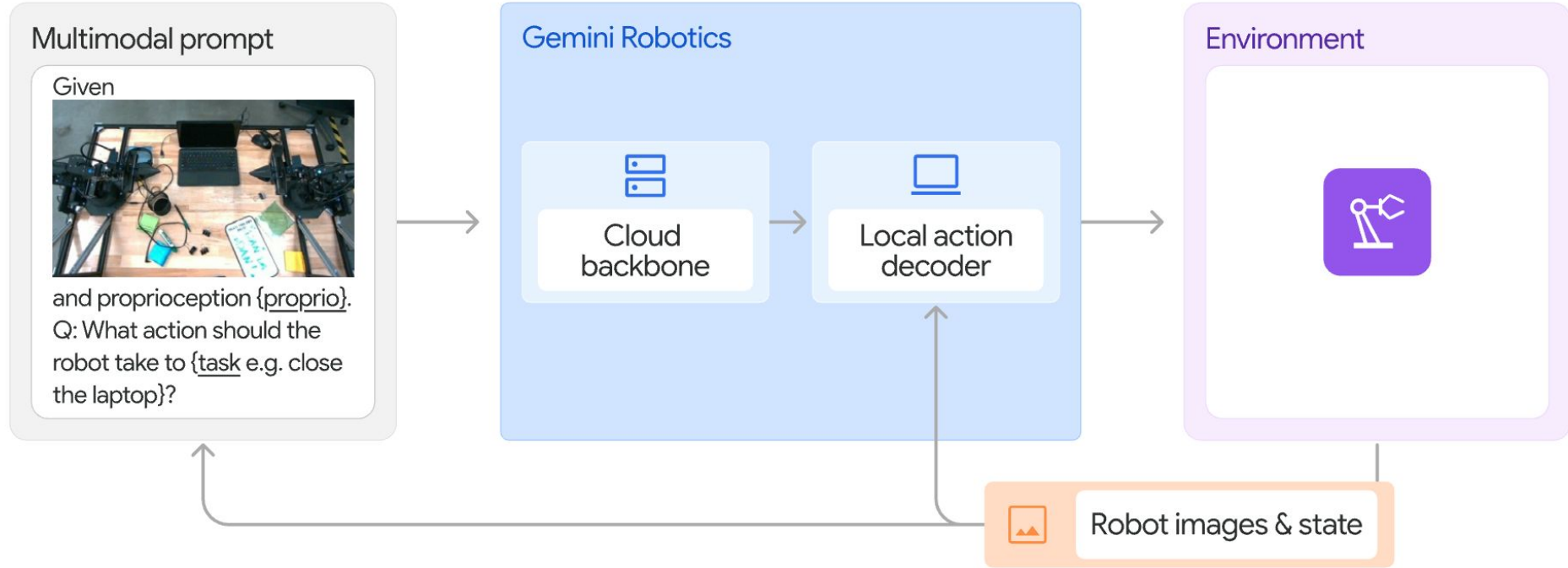


**Figure 12** | Overview of the perception and control APIs, and agentic orchestration during an episode. This system is used for zero-shot control.



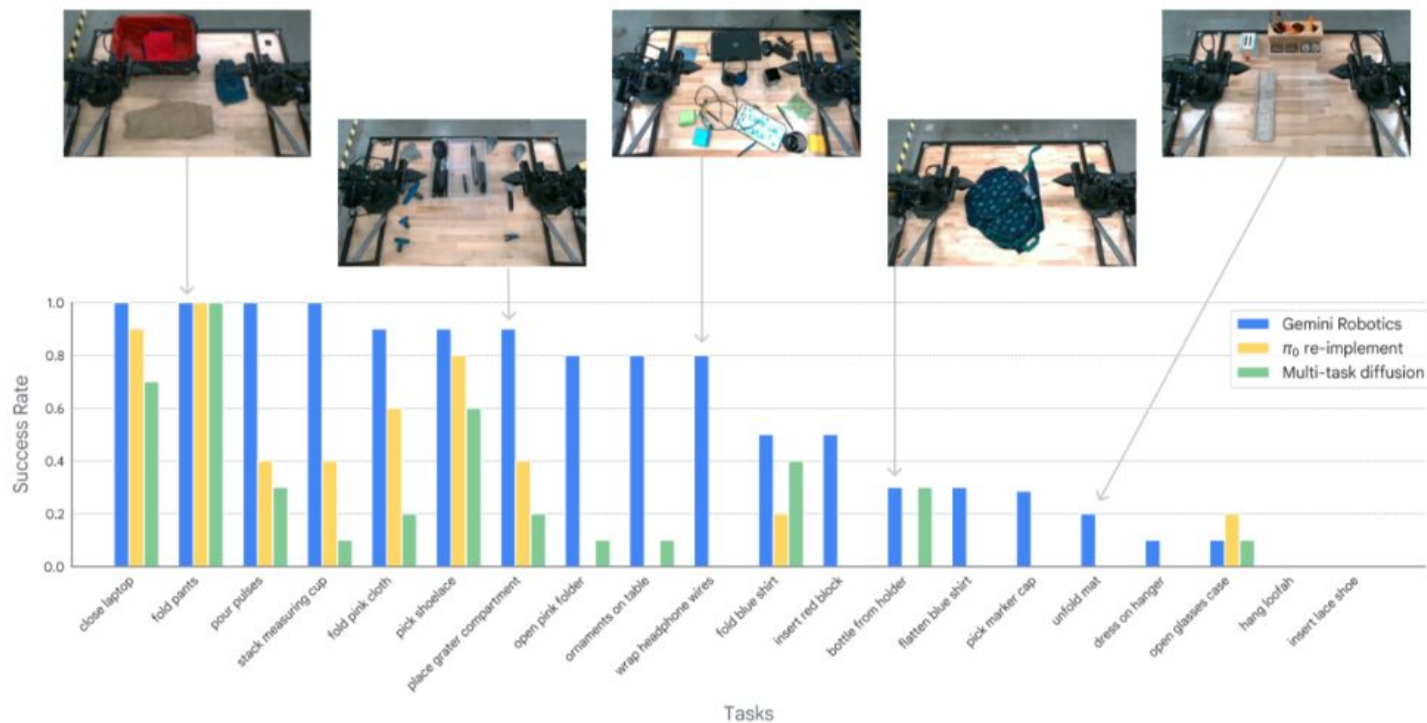
**Figure 13** | Overview of few-shot in-context learning pipeline. Gemini can receive observations, language instructions and trajectories in the prompt, and generate new language reasoning and trajectories for unseen instances of the tasks.

# Gemini Robotics

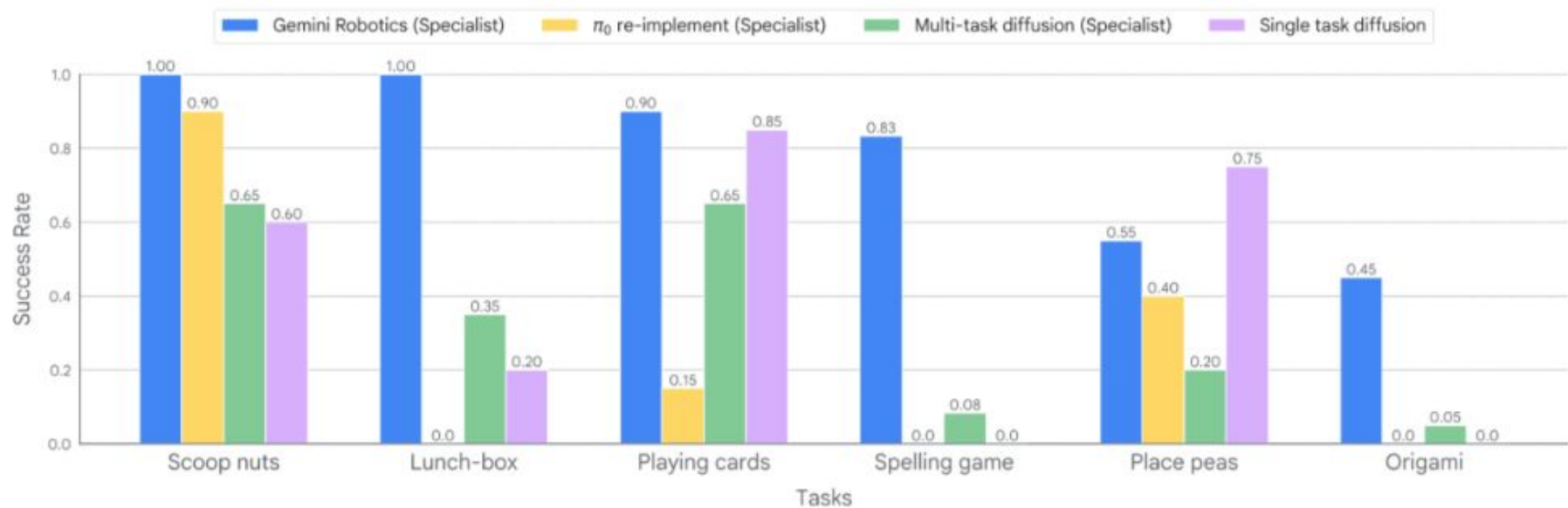


# Data

**Data.** We collected a large-scale teleoperated robot action dataset on a fleet of ALOHA 2 robots ([Team et al., 2024](#); [Zhao et al., 2025](#)) over 12 months, which consists of thousands of hours of real-world expert robot demonstrations. This dataset contains thousands of diverse tasks, covering scenarios with varied manipulation skills, objects, task difficulties, episode horizons, and dexterity requirements. The training data further includes non-action data such as web documents, code, multi-modal content (image, audio, video), and embodied reasoning and visual question answering data. This improves the



**Figure 16** | Gemini Robotics can solve a wide variety of tasks out of the box. We sample 20 tasks from the dataset, which require varying levels of dexterity, and evaluate our model and the baselines on them. Gemini Robotics significantly outperforms the baselines.



**Figure 23** | Performance on new, dexterous and long-horizon tasks after specialization. Gemini Robotics is the only model that can consistently solve the extremely challenging tasks like “Origami” and “Lunch-box”, achieving a 100% success rate on the latter, while baselines struggle with these tasks. While baselines are competitive on the easier tasks (such as “Scoop nuts”, “Playing cards” and “Place peas”), Gemini Robotics is the only successful method at the spelling game, accurately spelling printed picture cards and even achieving over 60% accuracy with hand-drawn sketches (which are never seen in training).

- **Gemini Robotics — ER (Embodied Reasoning) (2025)** — *VLM with embodied reasoning skills*
  - **Benchmark/Signals:** Trained/evaluated as a **VLM** with ER capabilities; introduces **ERQA (400** multi-choice Qs) covering spatial, trajectory, action reasoning, multi-view correspondence, etc. (for ER evaluation and supervision). Also leverages **internet-scale multimodal data** typical for Gemini 2.0.  
 Gemini Robotics  Gemini Robotics
- **Gemini Robotics — VLA (Control) (2025)** — *end-to-end VLA with reactive action chunks*
  - **Robot Action Data:** Large-scale **teleoperated** dataset on a fleet of **ALOHA-2** robots over **12 months, thousands of hours and thousands of diverse tasks** (varied skills, objects, horizons, dexterity).  Gemini Robotics  Gemini Robotics
  - **Additional Non-action Data:** **Web documents, code, image/audio/video, ER & VQA** data to bolster understanding/generalization.  Gemini Robotics
  - **Latency/Chunks:** Cloud **backbone** distilled from ER; local **action decoder** outputs **action chunks** (~250 ms end-to-end; 50 Hz effective control). (This is training/inference design tied to the action data above.)  Gemini Robotics  Gemini Robotics



| Evolution of AI in Robotics: From Symbolic Planning to Embodied Reasoning |                                                                                                                   |                                                                                                                                                                                                     |                                                                                                              |                                                                                                                                                      |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                  |
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| Work (year)                                                               | Problem / task focus                                                                                              | Main contribution                                                                                                                                                                                   | Method inputs / task spec                                                                                    | Outputs / action space                                                                                                                               | Key architectural choices                                                                                                                                                                                                                                                                    | Limitations / notes                                                                                                                                              |
| Natural Language Communication with Robots (≈2016; blocks-world style)    | Ground free-form human language to concrete manipulation plans in a simplified world                              | Shows that <b>unrestricted language</b> (not templates) can specify spatially complex goals; introduces datasets/evals for language→action grounding                                                | <b>Free-form NL</b> instructions; symbolic/visual world state (blocks, positions)                            | <b>Discrete, high-level skills</b> (e.g., pick/place/move block), not torque; low DoF abstraction                                                    | Semantic parsing / symbol grounding from NL → logical/graphical goals; rule-based or learned planner to execute                                                                                                                                                                              | Limited physical realism; sparse perception; <b>no low-level reactive control</b> ; struggles with ambiguous spatial language                                    |
| VIMA (2023)                                                               | General robot manipulation via <b>multimodal prompting</b> ; test zero-shot generalization                        | Reframes tasking as <b>interleaved text-and-image prompts</b> ; creates <b>VIMA-Bench</b> (L1→L4) for comp-gen generalization                                                                       | <b>Text + images / videos</b> as prompts specifying goals, constraints, exemplars; RGB observations from sim | <b>High-level parameterized skills</b> (pick/place/wipe) with pose parameters; <b>discrete primitive calls</b> , not torques                         | Encoder–decoder <b>Transformer</b> with <b>cross-attention</b> from observations/actions to prompt tokens; object-centric tokenization; large sim task suite                                                                                                                                 | Mostly simulation; <b>no high-frequency, low-level control</b> ; relies on pre-defined primitives; performance depends on prompt quality                         |
| π0 (pi-zero) (2024/2025)                                                  | <b>Reactive, fluent control</b> for diverse real-robot tasks across embodiments                                   | Introduces <b>continuous action distribution</b> via <b>conditional flow matching</b> and <b>action chunks</b> for smooth high-rate control; large cross-embodiment pretrain + dexterous post-train | <b>RGB (multi-view)</b> , <b>language, proprioception</b> ; optional task hinting                            | <b>Continuous action sequences</b> (e.g., 50 Hz <b>action chunks</b> ); low-level control over the robot's native <b>DoFs</b> (arms, grippers, etc.) | VLM backbone (e.g., Pali-Gemma-style) for perception/language + a dedicated <b>Action Expert</b> head trained with <b>flow-matching</b> ; <b>late fusion</b> of robot state; large, <b>cross-embodiment</b> dataset → <b>post-training</b> on long-horizon dexterity (e.g., laundry folding) | Data-hungry; safety/guardrails still needed; performance can depend on embodiment mismatch; training/serving compute heavier than discrete heads                 |
| Gemini Robotics – ER (Embodied Reasoning) (2025)                          | Build <b>fast visuo-spatial reasoning</b> skills robots need to act reactively (before control)                   | Adds ER skills (2D/3D detection, pointing, grasp/trajectory proposals, multi-view correspondence) + <b>ERQA</b> benchmark                                                                           | <b>Images / multi-view video</b> , language queries; sometimes depth                                         | <b>Perceptual/geometry outputs</b> : object detections, affordances, grasp/trajectory <b>proposals</b> ; not direct torques                          | Strengthened <b>VLM</b> with modules for spatial correspondence & grasp/trajectory prediction; multi-task ER training; ER-focused evaluation                                                                                                                                                 | Doesn't execute control by itself; reactive quality depends on downstream controller; may be sensitive to occlusion/precision geometry                           |
| Gemini Robotics – VLA (Control) (2025)                                    | <b>End-to-end vision-language-action</b> that produces <b>smooth, reactive movements</b> and adapts to new robots | Couples ER priors with a <b>low-level controller</b> for <b>high-frequency dexterous</b> manipulation; shows zero/few-shot generalization & <b>new-embodiment</b> transfer                          | <b>Open-vocab language</b> , RGB/video, robot state; optional demos                                          | <b>Low-level control commands</b> at the robot interface (continuous); supports varied <b>DoFs</b> ; emphasizes <b>reactivity</b> / smooth motion    | Family "stack": ER → VLA; policy that conditions on language + live perception; training on diverse manipulation corpora; specialization heads for dexterity/new embodiments                                                                                                                 | Hardware-/latency-dependent; requires curated action data; safety and failure recovery still active work; limited public detail on internals vs. research report |